

# Simplicial Morita equivalence and invariants

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# Morita theory in the simplicial setting

Many algebraic/homological features are Morita-invariant.

- Separability (DeMeyer-Ingraham 71')
- Global dimension (Lam 99') and GK dimension (McConnell-Robson 01')
- Twisted Calabi-Yau (Reyes-Rogalski 22')

Simplicial algebras carry additional homotopical data.

- Homotopy groups  $\pi_*(R)$

Comparison with other Morita theories (of dg/graded algebras).

- The Dold-Kan correspondence sends a simplicial algebra to a dg algebra, whose homology is a graded algebra.

# Studies in similar directions

Morita theory of monoids in suitable monoidal categories.

- Module-theoretical (Pareigis 78')
  - (Morita context) develop analogues of finitely generated projective modules and generators.
  - (Eilenberg-Watts theorem) characterize equivalences of module categories by tensoring with invertible bimodules.
- Functor-categorical (Lindner 74', Palmquist 75', Kelly 82', Borceux-Vitale 02', Lee-Park 24' etc)
  - View a monoid  $R$  as a category with a single object. Identify modules as presheaves on that category.
  - Characterize Morita equivalence as equivalence between Cauchy completion of functor categories.

# Main questions

Let  $\mathbb{k}$  be a commutative ring. By a simplicial algebra, we mean an associative unital simplicial  $\mathbb{k}$ -algebra.

1. Formulate the Morita theory for simplicial algebras using suitable simplicial analogues of finitely generated projective modules and generators.
2. How are homotopy groups of simplicial algebras related under simplicial Morita equivalence?
3. What are other some basic simplicial Morita invariants?

# Classical Morita theory

## Definition

Let  $R$  be a  $\mathbb{k}$ -algebra and  $M$  be a right  $R$ -module.

- $M$  is finitely generated projective if  $M$  is a direct summand of some  $R^{\oplus n}$ .
- $M$  is a generator if  $R$  is a direct summand of some  $M^{\oplus n'}$ .
- $M$  is an  $R$ -progenerator if  $M$  is a finitely generated projective generator.

## Definition

Let  $R$  and  $S$  be  $\mathbb{k}$ -algebras. We say  $R$  and  $S$  are Morita equivalent if  $\text{Mod } R$  is equivalent to  $\text{Mod } S$ .

# Morita equivalence

## Theorem

*Let  $R$  and  $S$  be  $\mathbb{k}$ -algebras. The following are equivalent.*

- *$R$  and  $S$  are Morita equivalent.*
- *There exists an  $R$ -progenerator  $M$  and an algebra isomorphism  $S \rightarrow \operatorname{Hom}_R(M, M)$ .*
- *There exists an invertible  $(R, S)$ -bimodule  $M$ .*

*In particular, any equivalence  $F : \operatorname{Mod} R \rightarrow \operatorname{Mod} S$  is naturally isomorphic to  $- \otimes_R M$  for some invertible  $(R, S)$ -bimodule  $M$ .*

## Example

- $R$  is Morita equivalent to  $M_n(R)$  for  $n \geq 1$ .
- $A$  is an Azumaya  $C$ -algebra (central separable over  $C$ ) if and only if  $A \otimes_C A^{op}$  is Morita equivalent to  $C$ .

# Simplicial objects

## Definition

A simplicial object  $X$  in a category  $\mathcal{C}$  is a collection of objects  $\{X_n\}_{n \geq 0}$  in  $\mathcal{C}$  with face maps  $d_i : X_n \rightarrow X_{n-1}$  and degeneracy maps  $s_i : X_n \rightarrow X_{n+1}$  being morphisms in  $\mathcal{C}$  for  $0 \leq i \leq n$  such that the following simplicial identities hold

$$\begin{aligned} d_i d_j &= d_{j-1} d_i & i < j, \\ s_i s_j &= s_{j+1} s_i & i \leq j. \end{aligned} \quad d_i s_j = \begin{cases} s_{j-1} d_i & i < j, \\ \text{id} & i = j \text{ or } i = j + 1, \\ s_j d_{i-1} & i > j + 1. \end{cases}$$

$$\cdots \rightrightarrows X_2 \rightrightarrows X_1 \rightrightarrows X_0$$

# Examples of simplicial objects

## Remark

A simplicial object in  $\mathcal{C}$  is precisely a functor  $\Delta^{op} \rightarrow \mathcal{C}$  where  $\Delta$  consists of finite totally ordered sets  $[n] := \{0, 1, \dots, n\}$  and non-strict order-preserving functions from  $[k] \rightarrow [n]$ .

## Example

The commutative ring  $\mathbb{k}$  can be viewed as a constant simplicial ring which is  $\mathbb{k}$  degreewise and face/degeneracy maps are identity maps.

## Example

The representable functor  $\Delta^n = \text{Hom}_{\Delta}(-, [n])$  is a simplicial set and  $\mathbb{k}\Delta^n$  is the simplicial  $\mathbb{k}$ -module freely generated by  $\Delta^n$ .

# Simplicial algebras and simplicial modules

## Definition

- A simplicial  $\mathbb{k}$ -module  $V$  is a simplicial object in  $\mathbb{k}\text{Mod}$ .
  - The category of simplicial  $\mathbb{k}$ -modules is a closed symmetric monoidal category (Schwede-Shipley 00').
- A simplicial algebra  $R$  is a simplicial object in  $\mathbb{k}\text{Alg}$ .
  - Equivalently, a simplicial algebra  $R$  is a monoid in the category of simplicial  $\mathbb{k}$ -modules.
- A right simplicial  $R$ -module is a simplicial  $\mathbb{k}$ -module  $M$  with a simplicial  $\mathbb{k}$ -module morphism  $M \otimes R \rightarrow M$  satisfying the usual associativity and unitality axioms.
  - Let  $\text{sMod } R$  denote the category of right simplicial  $R$ -modules.

# Example of a simplicial algebra

## Example

Consider the commutative ring  $\mathbb{k}$  as a constant simplicial algebra. Let  $M$  be simplicial  $\mathbb{k}$ -module.

The square-zero extension  $\mathbb{k} \ltimes M := \mathbb{k} \oplus M$  with multiplication

$$(a, x)(b, y) = (ab, ay + bx)$$

is a commutative simplicial algebra.

When  $M$  is not constant,  $\mathbb{k} \ltimes M$  is not constant.

When  $M \neq 0$ ,  $\mathbb{k} \ltimes M$  is not simple as  $M$  is an ideal.

# Simplicial enrichment

## Definition (Quillen 67')

Let  $M$  and  $N$  be right simplicial  $R$ -modules. The simplicial hom-set  $\mathbf{Hom}_R(M, N)$  is defined by

$$\mathbf{Hom}_R(M, N)_n = \mathrm{Hom}_{\mathbf{sMod} R}(\mathbb{k}\Delta^n \otimes M, N).$$

Such  $\mathbf{Hom}_R(M, N)$  is a simplicial  $\mathbb{k}$ -module.

## Proposition (Hovey 99')

*Let  $R$  be a simplicial algebra. Then  $\mathbf{sMod} R$  is  $\mathbb{k}\mathbf{sMod}$ -enriched.*

# Basic projectives and simplicial generators

## Definition (HSZZ 26')

Let  $R$  be a simplicial algebra and  $M$  be a right simplicial  $R$ -module. We say  $M$  is

- basic projective if  $M$  is a direct summand of some  $R^{\oplus n}$ .
- a simplicial generator if  $R$  is a direct summand of some  $M^{\oplus n'}$ .
- a simplicial  $R$ -progenerator if it is a basic projective generator.

## Proposition (HSZZ 26')

*(Dual Basis Lemma) The following are equivalent.*

- $P$  is a basic projective right simplicial  $R$ -module.
- *There are finitely many  $p_1, \dots, p_n \in P_0$  and  $q_1, \dots, q_n \in \text{Hom}_{\text{sMod } R}(P, R)$  such that  $p = \sum_{i=1}^n s_0^k(p_i)q_i(p)$  for all  $p \in P_k$  where  $k \geq 0$ .*

# Projectivity in enrichment

## Definition (HSZZ 26')

Let  $R$  be a simplicial algebra and  $P$  be a right simplicial  $R$ -module. We say  $P$  is

- projective if  $\mathrm{Hom}_R(P, -)$  is exact.
- simplicially projective if  $\mathbf{Hom}_R(P, -)$  is exact.
- basic projective if  $P$  is a direct summand of some  $R^{\oplus n}$
- finitely generated if there exist finite  $x_i \in P_{d_i}$  such that the smallest sub simplicial  $R$ -module of  $P$  containing  $\{x_i\}$  is  $P$ .

## Remark

We proved  $P$  is projective if and only if  $P$  is simplicially projective. Moreover, there exists  $P$  that is finitely generated projective but not basic projective.

# Simplicial Morita equivalence

## Definition

Let  $R$  and  $S$  be simplicial algebras. We say  $R$  and  $S$  are Morita equivalent if  $\mathbf{sMod} R$  and  $\mathbf{sMod} S$  are equivalent as  $\mathbb{k}\mathbf{sMod}$ -enriched categories.

## Definition

Let  $R$  be a simplicial algebra. For  $n \geq 1$ , the simplicial matrix algebra  $M_n(R)$  is

$$\Delta^{op} \xrightarrow{R} \mathbb{k} \mathbf{Alg} \xrightarrow{M_n(-)} \mathbb{k} \mathbf{Alg}$$

where  $M_n(-)$  is the usual matrix ring construction.

## Theorem (HSZZ 26')

*Let  $R$  and  $S$  be simplicial algebras. The following are equivalent.*

- (1)  $R$  and  $S$  are Morita equivalent.*
- (2) There exists a simplicial  $R$ -progenerator  $P$  such that  $S \cong \mathbf{Hom}_R(P, P)$  as simplicial algebras.*
- (3) There exists an invertible simplicial  $(R, S)$ -bimodule  $M$ .*
- (4) For some positive integer  $n$ ,  $S \cong eM_n(R)e$  as simplicial algebras for a full idempotent  $e \in M_n(R)_0$ .*

*In particular, any  $\mathbb{k}\mathbf{sMod}$ -enriched equivalence*

*$F : \mathbf{sMod} R \rightarrow \mathbf{sMod} S$  is naturally isomorphic to  $- \otimes_R M$  for some invertible simplicial  $(R, S)$ -bimodule  $M$ .*

## Remark

*(1), (2), and (3) are specializations of Lindner 74', Palmquist 75', Pareigis 78', Kelly 82', Borceux-Vitale 02', Lee-Park 24' etc.*

# Homotopy groups are graded Morita equivalent

## Definition

The  $n$ th homotopy group of a simplicial  $\mathbb{k}$ -module  $V$  is  $\pi_n(V)$ .

## Theorem (HSZZ 26')

*Suppose  $R$  and  $S$  are Morita equivalent simplicial algebras.*

- (1) If  $\pi_0(R) = 0$ , then  $\pi_*(R) = 0 = \pi_*(S)$ .*
- (2) If  $\pi_0(R) \neq 0$ , then  $\pi_*(R)$  and  $\pi_*(S)$  are Morita equivalent as unital graded algebras.*

## Some basic invariants

### Definition (HSZZ 26')

Let  $R$  be a simplicial algebra. Define the simplicial center  $Z(R)$  by  $Z(R)_n = \{r \in Z(R_n) \mid \theta_R^*(r) \in Z(R_k), \forall \theta : [k] \rightarrow [n] \text{ in } \Delta\}$

### Definition (HSZZ 26')

Let  $R$  be a simplicial algebra. The Picard group  $\text{Pic}(R)$  is the group of isomorphism classes of invertible simplicial  $R$ -bimodules with group operation  $\otimes_R$ . The identity of  $\text{Pic}(R)$  is  $[R]$ .

# Morita Invariants

## Proposition (HSZZ 26')

*Let  $R$  and  $S$  be Morita equivalent simplicial algebras.*

- (1)  $Z(R) \cong Z(S)$ .
- (2)  $\text{Pic}(R) \cong \text{Pic}(S)$ .
- (3)  $\pi_n(R) = 0$  if and only if  $\pi_n(S) = 0$ .
- (4)  $R$  is isomorphic to a constant simplicial algebra if and only if so is  $S$ .

# Questions for future studies

We have characterized the separable simplicial algebras and the Azumaya simplicial algebras.

- Is simplicial separability invariant under simplicial Morita equivalence?

Twisted Calabi-Yau algebras of dimension 0 are equivalent to separable algebras (Reyes-Rogalski 22').

- How to generalize the twisted Calabi-Yau condition to simplicial algebras?
- Is the twisted Calabi-Yau condition simplicial Morita invariant?
- Are twisted Calabi-Yau simplicial algebras of dimension 0 equivalent to separable simplicial algebras?

*Thank you!*

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